POKHARA UNIVERSITY

Level: Bachelor | Semester: Fall | Year : 2023
Programme: BE | Full Marks:100
Course: Programming in C (New) | Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) What is a programming language? Explain all the generations of programming language. 7

   b) Explain importance of algorithm and flowchart. Draw a flowchart to check if given year is leap year or not. 8

2. a) What are keywords? Explain types of operators based on number of operands with example program. 7

   b) Electricity board charges according to following rules. 8

   For first 100 units----------> Rs 100 (minimum charge)
   For next 50 units ----------> Rs 8.5 per unit
   For next 100 units --------- > Rs 9.5 per unit
   For beyond 250 units ------> Rs 10.5 per unit

   Additional charge:13% tax and Rs 100 for maintenance.
   Write a program to input total units consumed and display total charge.

3. a) What is control statement? Explain break, continue and goto control statements with example program. 7

   OR

   Explain for loop. WAP to convert decimal numbers to binary bits.

   b) What is string? Explain any 7 predefined string handing functions with example. 8

4. a) Write a program to input n elements in a 1D array and check is a given number is present in the array or not. If present, display the position of the number in array. 7

   b) Define function with syntax and mention its advantages in POP. Explain different storage classes in C programming. 8

5. a) What is recursive function? Write a program to find power of given number using recursion. 7

   b) What is dynamic memory allocation? Explain functions used in dynamic memory allocation with example. 8

6. a) Create structure named Student. Student has attributes name, address, date of birth and marks in percentage. Input the records for 'n' students and display information of the students who have scored above 75 percentage. 7

   b) Why is file handling necessary in C programming? Write a program to input name, address, faculty, program, and GPA (in maximum 4.0) of 500 students and store them in 'RESULT.DAT' data file and display the records of those students whose faculty is 'Engineering and GPA >3.5'. 8

   OR

   Write a program to create a structure Teacher with attributes name, address, faculty and salary. Input records of 50 Teachers and store them in "Teacher.txt" file and display the records of teachers whose address is "Gorkha"

7. Write short notes on: (**Any two**) 2×5

   a) Data types
   b) Call by reference using pointer
   c) self-referential structure